Metaheuristics

Academic year 2025/2026

Assigment 1
(3 weeks)

Problem statement

En random:Intentar que todos los segmentos tengan la misma chance de obtener un nº alto (Random points --> reoorder)

This first assignment is aimed to the development and application of heuristic and metaheuristic methods for problem solving. Students must apply, using Python, two heuristics (Random Search and Hill Climbing) and one metaheuristic (Simulated Annealing) to solve the proposed problem. Students must also submit a formal report justifying which of the implemented methods is the most suitable for the proposed problem.

Para generar vecionos variamos en un porcentage de variabiliadad el valor de los separadores. EJ) 1% 200ptos es 2ut.
Si es decimal aproximamos

It is mandatory to consider at least the following elements to justify the answer:

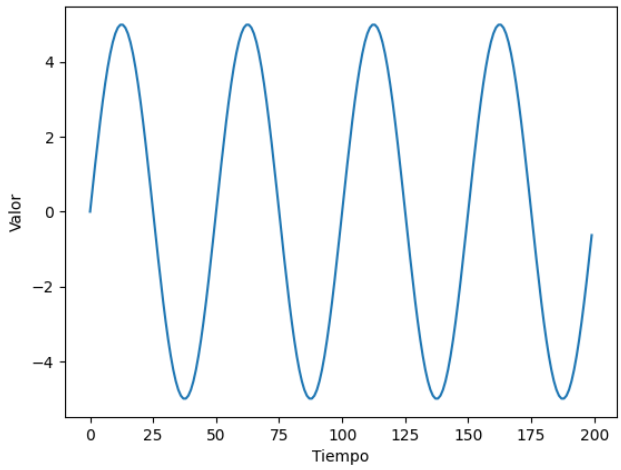
- Accuracy of the results → Mejor Fitness
- Variability of the results
- Computational time

Problem

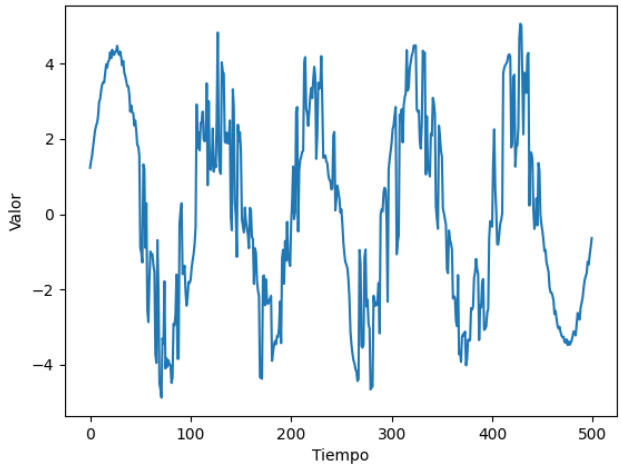
The goal is to segment a time series into a number k of segments. The width of each segment (number of points included in the segment) is variable and unknown. For each segment, a linear regression must be fitted using all the points within that segment. In other words, the objective is to represent a line such that the distance from the points to that line is minimized for all points in the segment. The aim is to determine the k segments (the bounds of each segment) that minimize the average of the mean squared errors (the average of the squared errors of each segment).

Consider the following time series and k values:

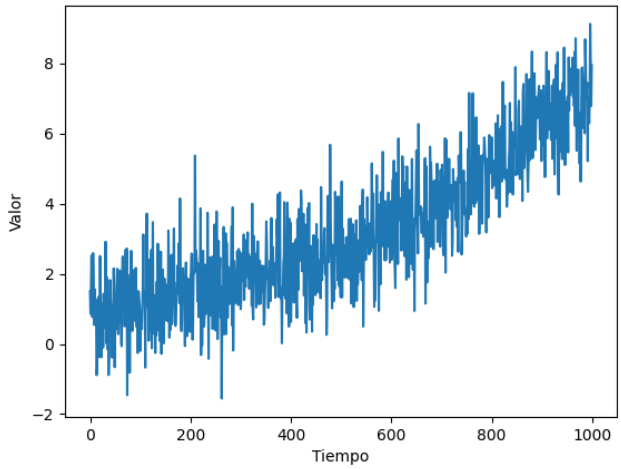
- TS1. K = 9



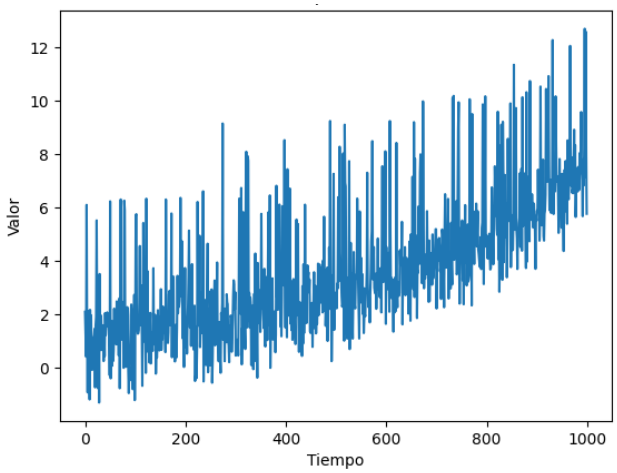
- TS2. K = 10



- TS3. K = 20



- TS4. K = 50



Deliverables

Students must submit:

- A PDF report.
- Python source code of the algorithms. Google Colab may be used; in that case, the public link to access and execute the code must be submitted.
- Readme.md que explique como ejecutarlo

Evaluation Explicar cada decisión

- Comparison and Analysis (70%). Justification of the methods (suitability for the problem, strengths, weaknesses, etc.), evaluation of algorithm performance, and discussion of the obtained results, following the points outlined in this assignment.
- Formal-Library Report and Code Presentation (30%). Formal-style documentation, with clear explanations of the algorithms and clarity in the comparative analysis. Well-commented code.

Normalmente + 10/15 pág